

Physics

1. Derivation of Lagrange's equations

1. Consider a system of particles: $k = 1, \dots, N_P$. Let $r^{(k)}$ be the position of the k^{th} particle relative to an inertial reference frame. Let $f^{(k)}$ be the resultant force acting on the mass m_k of the k^{th} particle.
2. Then, from Newton's Second Law:
 1. $f^{(k)} = m_k \ddot{r}^{(k)}$, $k = 1, \dots, N_P$
 2. or $f^{(k)} - m_k \ddot{r}^{(k)} = 0$, $k = 1, \dots, N_P$
3. Let $\delta r^{(k)}$ be a 'virtual' displacement that is compatible with all constraints on the motion of the k^{th} particle.

References:

1. Dynamics, Brown university.